



AURA Analyst

AI-Powered Autonomous Data Analysis & Forecasting Platform

"Transforming Raw Data into Actionable Insights using Generative AI"

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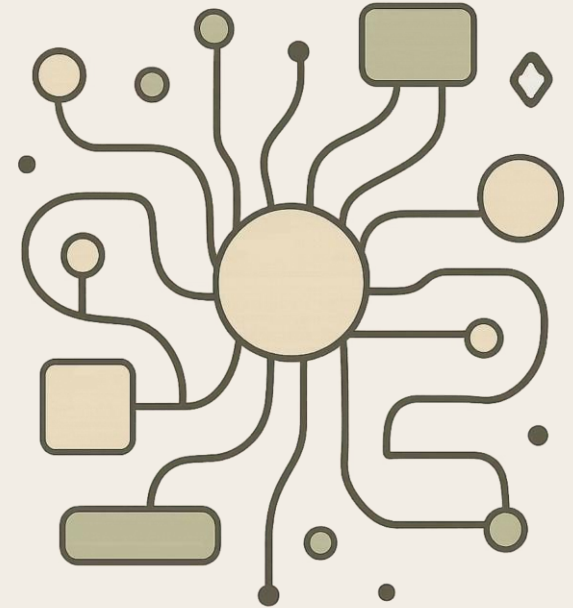
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Introduction & Problem Statement

The Challenge

Organizations face rapid data growth, yet existing tools like Excel and Tableau require technical expertise and hide critical insights.

Key Problems

- Coding knowledge required
- Time-consuming manual analysis
- Hidden insights difficult to discover
- Limited tool capabilities

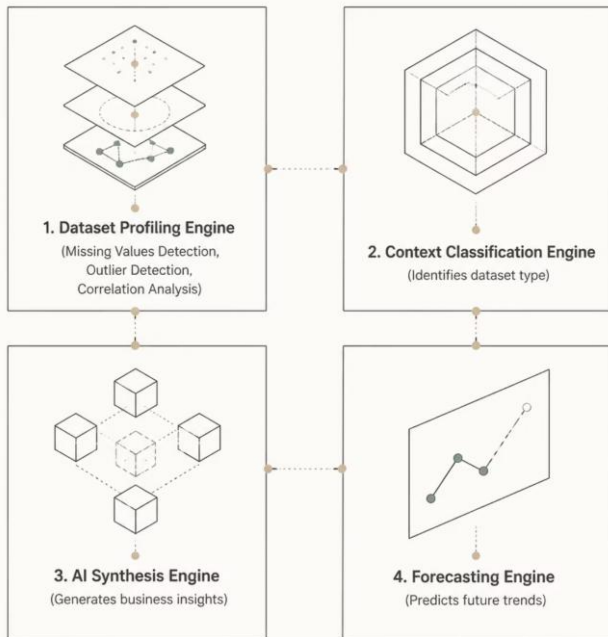
Proposed Solution

AURA Analyst enables users to upload datasets and receive AI-generated insights, visualizations, forecasts, and answers using natural language.



Proposed System

Four intelligent engines power the AURA Analyst platform.



Dataset Profiling Engine

Detects missing values, outliers, and performs correlation analysis automatically.

Context Classification Engine

Identifies dataset type to tailor the analysis approach.

AI Synthesis Engine

Generates actionable business insights from processed data.

Forecasting Engine

Predicts future trends using AI-driven models.

Objectives & Requirements

Core Objectives

- Automated Data Analysis
- AI-Generated Insights
- Natural Language Querying
- Forecast Generation
- Interactive Dashboard
- Multi-file Analysis

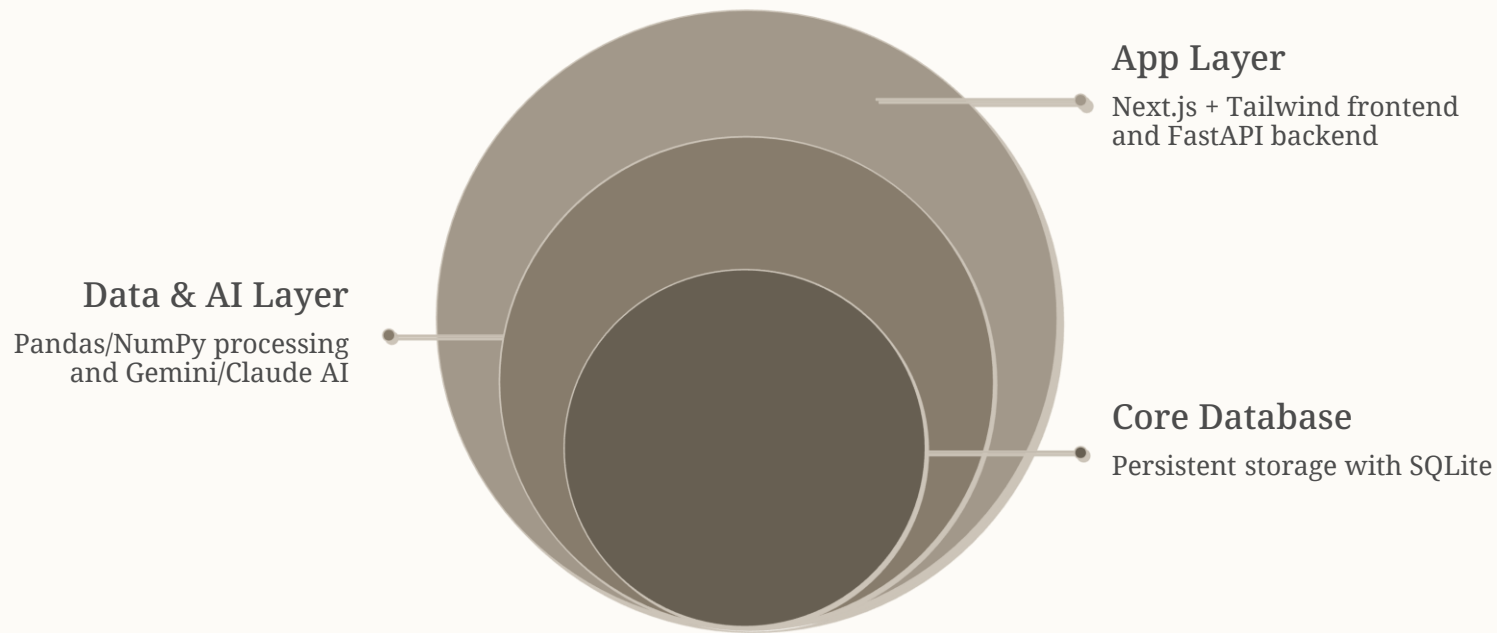
Functional Requirements

- CSV Upload & Dataset Profiling
- AI Analysis & Forecasting
- Q&A Interface

Non-Functional Requirements

- Fast Response Time
- Scalability & Reliability
- Security

System Architecture



A clean layered architecture ensures separation of concerns — from the Next.js frontend through FastAPI backend, AI models, data processing, and persistent storage in SQLite.

System Design & Workflow

01

User Uploads Dataset

02

Dataset Profiling

03

Context Classification

04

AI Insight Generation

05

Forecast Generation

06

Dashboard Visualization

07

Natural Language Q&A

Each step is fully automated — from dataset upload through intelligent analysis to interactive Q&A — delivering end-to-end autonomous data intelligence.

Technologies Used



Frontend

Next.js, React, TypeScript, Tailwind CSS



AI Models

Claude Sonnet, Claude Opus



Visualization

Plotly



Backend

Python, FastAPI



Data Processing

Pandas, NumPy



Database

SQLite

Implementation & Testing



Development Methodology

Agile Development with iterative sprints and continuous feedback.

Testing Types

- Unit Testing
- Integration Testing
- System Testing

Test Results

✓ Successful CSV Upload

✓ Accurate Profiling

✓ Correct Forecast
Generation

✓ Functional AI Q&A

Results & Achievements

Key Achievements

Performance Highlights

- Analysis Time < **30 Seconds**
- Q&A Response Time **2–5 Seconds**
- Multi-file Analysis **Supported**

✓ Automated Dataset
Analysis

✓ AI-Generated
Insights

✓ Forecasting
Capability

✓ Interactive
Visualizations

✓ Natural Language
Interaction

Conclusion & Future Scope

Conclusion

AURA Analyst successfully automates data analysis, insight generation, and forecasting using Generative AI. It reduces dependency on technical expertise and enhances decision-making efficiency.

Future Enhancements

- Real-Time Data Integration
- Multi-Language Support
- Advanced Predictive Models
- Cloud Deployment
- API Integration

Thank You

